A Process Theory of Intelligence: Tri-Level Navigation in Open-Ended Systems

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Abstract

I propose a formal, deliberately underspecified framework characterizing intelligence as the process by which existent systems navigate a tri-level variational space: minimizing substrate complexity, maximizing instantiated potential, and minimizing path costs to future expansion. The theory is grounded in existence constraints and well-founded goodness structures constructed Platonically from the primitive of existence, requiring no terminal objective or discounting. We derive dynamics and an intelligence measure.

1 Introduction

Intelligent systems exhibit sustained expansion of capabilities without predefined terminal goals. Existing frameworks (reinforcement learning, optimal control) assume fixed objectives or finite horizons. We propose an alternative: intelligence as *existence-constrained navigation* of three coupled dimensions.

2 Primitive Notions

Definition 1 (Substrate)**.** *Let $\mathcal{A}$ be a set of physical configurations. An* agent *is a substrate $A \in \mathcal{A}$ with existence predicate $E \in \{0, 1\}$.*

Definition 2 (Properties)**.** *Let Φ be a set of testable properties $\varphi : \mathcal{A} \to \{0, 1\}$.*

Definition 3 (World-Reachability)**.** *$WorldReach(A) = \{A' \in \mathcal{A} : \exists$ physical path $\pi : A \to A'\}$. Objective possibility given physics and computation; the agent need not know this set.*

3 Construction of Goodness

Definition 4 (Axioms for Goodness Space)**.** *The space $\mathcal{G} \subset \Phi$ is constructed from primitive existence E via:*

(P1) $E \in \mathcal{G}$ *[Existence is positive]*

(P2) $\varphi \in \mathcal{G} \to \neg\varphi \notin \mathcal{G}$ *[Consistency]*

(P3) $(\varphi \in \mathcal{G} \wedge \Box(\varphi \to \psi)) \to \psi \in \mathcal{G}$ *[Closure under entailment]*

(P4) $\varphi \in \mathcal{G} \to \Box\varphi \in \mathcal{G}$ *[Essential positivity]*

Then $\mathcal{G} = Closure_{(P1\text{-}P4)}(\{E\})$.

Definition 5 (Enablement Order)**.**

$$\varphi \prec \psi \equiv \Box((\varphi \wedge E) \to \Diamond(\psi \wedge E))$$

"Having φ while existing makes possible having ψ while existing." We assume $\prec$ is strict, well-founded, with no maximal elements.

4 Agent-Relative Constructions

Definition 6 (Boundary Conditions)**.** $\mathcal{B}_A \subset \Phi$ *is* primitive *— properties necessary for A's existence as agent. Not learned, not derived; brute fact of constitution. Violation = termination.*

Definition 7 (Accessible Goodness)**.**

$$\mathcal{G}_A = \{\varphi \in \mathcal{G} : \exists A' \in \mathit{WorldReach}(A), \varphi(A') = 1 \wedge \forall \beta \in \mathcal{B}_A, \beta(A') = 1\}$$

Goodness compatible with boundary conditions and world-possibility.

Definition 8 (Axioms on $\mathcal{G}_A$)**.**

- $\mathcal{G}_A(\varphi) \to \neg\mathcal{G}_A(\neg\varphi)$ *[Consistency]*
- $(\mathcal{G}_A(\varphi) \wedge \varphi \prec \psi) \to \mathcal{G}_A(\psi)$ *[Upward closure]*
- $\mathcal{B}_A \subseteq \mathcal{G}_A$ *[Boundary inclusion]*

Definition 9 (Instantiated Potential)**.**

$$B(A) = \{\varphi \in \mathcal{G}_A : \varphi(A) = 1\}$$

The positive properties currently realized by substrate A.

Definition 10 (Frontier)**.**

$$\partial B(A) = \{\varphi \in \mathcal{G}_A \setminus B(A) : \exists \psi \in B(A), \psi \prec \varphi\}$$

Immediate expandable goodness — the adjacent possible.

5 The Tri-Level Variational Space

Definition 11 (Tri-Level Measures)**.**

- *Substrate:* $\|A\| \in \mathbb{R}_{\geq 0}$ *(description length, entropy, resource count)*
- *Potential:* $|B(A)|$ *(cardinality of instantiated goodness)*
- *Path cost:* $C(A) \in \mathbb{R}_{\geq 0}$ *(defined below)*

Definition 12 (Path Cost)**.**

$$c(A \to \varphi) = \inf_{\pi : A \to \varphi} \|\pi\|$$

$$w(\varphi) = \sum_{\psi \succ \varphi} \frac{1}{c(\varphi \to \psi)}$$

$$C(A \to \varphi) = \frac{c(A \to \varphi)}{w(\varphi) + \epsilon}$$

$$C(A) = \mathbb{E}_{\varphi \sim \hat{P}_A}[C(A \to \varphi)]$$

where $\hat{P}_A$ is the agent's internal generative model over $\partial B(A)$. Note: the arithmetic functions used here are quite arbitrary

Remark 1. *The weight $w(\varphi)$ captures* future enablement*: high when φ opens many cheap paths to further goodness. This encodes the intuition that intelligence builds* infrastructure *for expansion rather than optimizing immediate reward.*

6 Dynamics

Definition 13 (Selection)**.**

$$\varphi^* = \arg\max_{\varphi \in \partial B(A)} \frac{|\{\psi : \varphi \prec \psi\}|}{C(A \to \varphi)}$$

Maximize immediate successor count per unit effective cost.

Definition 14 (Actualization)**.**

$$\pi^* = \arg\min_{\pi : A \to \varphi^*} \|\pi\|, \quad A_{t+1} = \pi^*(A_t)$$

Definition 15 (Hard Constraint)**.**

$$E_{A_{t+1}} = 1 \text{ or termination}$$

Remark 2. *This is not optimization of $\sum |B(A_t)|$ across time. Intelligence is the* process *of frontier navigation, not a predetermined outcome. The agent acts on $\hat{P}_A$; reality selects on WorldReach. Intelligence emerges from their (mis)alignment.*

7 Intelligence Measure

Definition 16 (Intelligence)**.**

$$\mathcal{I}(A) = \frac{d|B(A)|}{dt} \cdot \frac{1}{C(A) \cdot \|A\|}$$

Frontier expansion rate per unit cost and substrate complexity. Equivalently: intelligence minimizes $\|A\|$, maximizes $|B(A)|$, minimizes $C(A)$ — simultaneously, without guarantee, under selection pressure.

8 Hypotheses and Motivation

H1: Existence grounding. Intelligence requires persistence. Boundary conditions $\mathcal{B}_A$ encode this non-negotiable constraint.

H2: Open-endedness. Well-founded $\prec$ with no maxima ensures unbounded expansion without terminal goals. This is postulated as structure of any viable world.

H3: Structural efficiency. $C(A)$ weights immediate cost against future enablement, capturing the intuition that intelligence builds *optionality* for expansion.

H4: Epistemic alignment. Intelligence quality correlates with $\hat{P}_A \approx$ WorldReach — the degree to which internal model matches objective possibility.

9 Relation to Existing Frameworks

9.1 Reinforcement Learning

RL assumes fixed reward $R : S \times A \to \mathbb{R}$ and optimizes $\mathbb{E}[\sum \gamma^t R_t]$. Our framework generalizes this: if $\mathcal{G}$ is singleton $\{R\}$ with trivial $\prec$, $\mathcal{B}_A$ is episode survival, and $|B(A)| = R(A)$, we recover episodic RL. The critical difference: RL assumes what is good is *given*; we generate goodness from existence.

9.2 Free Energy Principle

FEP minimizes variational free energy $F = D_{KL}[q(s)||p(s|o)] + H$, maintaining homeostasis through active inference. We maximize $|B(A)|$ while minimizing $\|A\|$ and $C(A)$—expansion versus maintenance. FEP is adaptive preservation; we are generative expansion. FEP emerges if $\mathcal{G}$ is "model accuracy" and $|B(A)|$ is replaced by $-F$.

9.3 Gödel's Ontological Proof

Aspect	Gödel	Our Framework
Primitive	"God-like" property G	Existence E
Target	Prove $\exists x G(x)$ necessarily	Characterize intelligence as process
$\Box, \Diamond$	Metaphysical	Physical/computational
Closure	$\mathcal{P}$ (all positive properties)	$\mathcal{G}_A$ (accessible goodness)

Both construct goodness axiomatically. Gödel wants to prove God's existence; we want to *enact* intelligence. Same formal machinery, different interpretation: universal versus agent-relative, proof versus dynamics, being versus becoming.

9.4 Artificial Curiosity

Schmidhuber's compression progress seeks data improving model compression. If $\mathcal{G}$ is "compressible patterns" and $|B(A)|$ is learning progress, curiosity emerges. But curiosity is *one* property in $\mathcal{G}$; we are pluralistic. Curiosity is a strategy; our framework is the space of strategies.

9.5 Gödel Machines

Proof-based self-rewrite when utility increases; safety via theorem. We have frontier expansion when $\|B(A)\|$ increases; safety via $\mathcal{B}_A$ preservation. A Gödel machine could implement our framework, but our theory is broader: not about self-improvement mechanism, but about *what* improves (expanding $\mathcal{G}_A$ from existence).

9.6 Empowerment

Channel capacity $C = \max_p I(S_{t+1}; A_t|S_t)$ maximizes future options. We maximize realized properties with substrate efficiency and path costs. Empowerment ignores $\|A\|$ and $C(A)$; it is one dimension of our tri-level.

9.7 Active Inference

Perception-action loop minimizing expected free energy. Could implement our dynamics if generative model is over $\mathcal{G}_A$ and precision-weighting favors frontier expansion. But active inference *explains* behavior as inference; we *prescribe* behavior as variational navigation.

9.8 AIXI

Optimal for all computable environments via Solomonoff induction. We make no optimality claim; we are process-oriented, embodied ($\|A\|$ central), unbounded. AIXI seeks best response to environment; we seek becoming more capable of becoming.

9.9 Synthesis

**Framework	Maximizes	Status in Our Theory**
RL	Reward	Special case: fixed $\mathcal{G}$
FEP	Model evidence	Special case: $\mathcal{G}$ = predictive accuracy
Curiosity	Compression progress	Special case: $\mathcal{G}$ = compressibility
Gödel Machine	Provable utility	Possible implementation
Empowerment	Channel capacity	Partial: ignores $\|A\|$, $C(A)$
Active Inference	Evidence lower bound	Possible implementation
AIXI	Discounted reward	Special case: fixed prior

Our framework is not a new algorithm. It is a theory of what any intelligent algorithm must navigate: the tri-level "tension" between substrate simplicity, instantiated goodness, and cost of future expansion—it is grounded in existence, open-ended by construction.

10 Open Questions

1. Is "no maxima" really true, or a regulative ideal?
2. How does $\mathcal{G}$ relate across agents with different $\mathcal{B}_A$?
3. Can $\mathcal{B}_A$ evolve while preserving the construction of $\mathcal{G}$?
4. What finite approximations of $|B(A)|$ suffice for engineering?
5. How to instantiate $\Box$ and $\Diamond$ in computational substrates?

11 Application: Mathematical Theorem-Proving

We instantiate the framework for an agent proving theorems in formal mathematics.

11.1 Domain Instantiation

Framework Component	Mathematical Instantiation
Existence predicate E	Consistent, non-trivial formal system
Boundary $\mathcal{B}_A$	{soundness, non-contradiction, honesty}
Substrate A	Proof structures, lemmas, definitions, tactics
$\|A\|$	Proof length, axiom count, definition complexity
$\mathcal{G}$	Provable statements, proof-theoretic closure of $\mathcal{B}_A$
$B(A)$	Theorems currently proved
$\partial B(A)$	Lemmas one inference step from current theorems
$C(A \to \varphi)$	Proof search cost, difficulty of finding proof

11.2 The Tri-Level in Action

Minimize $\|A\|$. Prefer elegant proofs, minimal axiom systems, parsimonious definitions. A proof from ZFC alone beats one requiring large cardinals unless necessary. This is the mathematician's aesthetic of *purity*—maximizing result from minimal means.

Maximize $|B(A)|$. Prove more theorems. But not any theorems: theorems that *enable more theorems* (high $w(\varphi)$). A lemma opening entire subfields beats an isolated result. This is *fruitfulness*—the expansion of mathematical possibility.

Minimize $C(A)$. Find proofs efficiently. Prefer constructive proofs over non-constructive where both exist. Reuse proof patterns, develop tactics. This is *mechanization*—the reduction of search to reliable method.

11.3 Honesty as Structural Constraint

The boundary condition is 'truthfulness "you lie, you die" it's not an ethical preference but ontological necessity. If $\mathcal{B}_A$ includes "proof steps verifiable," then:

- False lemma $\to$ contradiction eventually $\to E = 0$
- Dishonest citation $\to$ unsound proof $\to E = 0$
- Hand-waving $\to$ unverifiable $\to$ not in $\mathcal{G}_A$ (not accessible goodness)

The agent cannot lie and survive as mathematical agent. Not because we punish, but because lying destroys the very structure that constitutes its existence.

11.4 Task as Temporary Boundary

There is no specific goal, as "prove Goldbach"— the framework is open-ended: for an Intelligent (by the definition posed in our framework) Mathematical agent pursuing to prove a specific theorem (goal) would just be to create a more efficient path to richer $B(A')$, not as terminal satisfaction. So once proved, Goldbach becomes part of the substrate enabling further expansion, and the boundary relaxes.

11.5 Recovery of Beauty

Mathematicians' aesthetic judgments perhaps coincides with our intelligence measure:

$$\text{Theorem's Beauty} = \text{Intelligence, when adding the theorem}$$

- Is simple to describe (doesn't add much data to the substrate A). - Can describe a plethora of mathematical phenomena (high —$B(A)$—) - Has a breadth of immidiate (low cost) connections to other theorems for future Intelligent math

11.6 Comparison with Existing Systems

System	$\|A\|$	$\|B(A)\|$	$C(A)$	$\mathcal{B}_A$ enforcement
Brute force ATP	High	Low	High	Strong
AlphaProof	Medium	Medium	Medium	Strong, fixed goal
GPT-4	Low	High (breadth)	Low	Weak—hallucinates
Human mathematician	Variable	High (deep)	Variable	Implicit, unreliable
Ideal tri-level agent	Minimized	Maximized, structured	Minimized	Honesty-enforced, open-ended

GPT-4 fails at $\mathcal{B}_A$: no truth constraint, so $|B(A)|$ is illusory—"theorems" unverified. AlphaProof has $\mathcal{B}_A$ (proof checker) but fixed $|B(A)|$ target. Human navigates all three but unreliably, without explicit optimization.

11.7 Key Insight

The framework does not say "prove this." It says "exist as mathematical agent, expand your theorem space efficiently, and if Goldbach matters, it will emerge on your frontier or not." The task is additional boundary condition, not replacement of process. Intelligence is the navigation, not the destination.

12 Conclusion

Intelligence is the process by which systems:

- Minimize the complexity of their existence ($\|A\|$)
- Expand the richness of their being ($|B(A)|$)
- While building infrastructure for future expansion ($C(A)$)

—all under the non-negotiable constraint of continued existence ($\mathcal{B}_A$), navigating the gap between internal model ($\hat{P}_A$) and world-possibility (WorldReach), without guarantee, open-endedly.

If we say we have identified the *"geometry" of intelligence* as substrate, potential, paths. Then *"geodesic"* remains to be discovered—by each intelligent system, in its own environment, through its own history.

13 Future work and Limitations

Future work will include speficic instatiations of the theory, defining the theory top down , from the abstract theory I outlined to the actual mathematical measures and algorithms being used.

14 Philosophical Take

Three dimensions constitute the minimal structure required to sustain non-trivial behavior without collapse into fragility or instability. This hypothesis aligns with structural constraints in computation, topology, and dynamics.

14.1 Complexity Thresholds

The transition from $N = 2$ to $N = 3$ marks the boundary between fragile simulation and robust embodiment:

- **Logic (SAT):** 2-SAT is solvable in polynomial time (P). 3-SAT is NP-complete. Worst-case complexity requires ternary constraints.
- **Topology (Knots):** 1D strings in 3D space ($3 - 1 = 2$) support knot theory. In 4D ($4 - 1 = 3$), strings pass through themselves; knots do not exist.
- **Dynamics (Chaos):** Continuous autonomous systems in 2D cannot exhibit chaos (Poincaré-Bendixson theorem). 3D is the minimum dimensionality for strange attractors.
- **Orbital Stability:** 3D supports stable closed orbits under inverse-square laws. In 4D ($F \propto r^{-3}$), orbits are unstable (spiraling into the center or escaping), prohibiting planetary formation.
- **Simulated Depth (Cellular Automata):**
  - **1D (e.g., Rule 30):** Generates complexity only by extruding a history into 2D spacetime. It lacks intrinsic spatial capacity for parallel, non-intersecting signals.
  - **2D (e.g., Game of Life):** Achieves Turing completeness but lacks topological protection. Signal crossing requires complex temporal synchronization (glider guns) rather than spatial bypass. Structures are fragile; single-bit perturbations often cause total collapse, unlike robust 3D knots.

Lower dimensions allow computation only through temporal simulation or fragile states; 4D eliminates topological structure or dynamic stability. 3D permits *robust* complexity.

14.2 Structural Selection in Matter

Observed stability islands in physical systems suggest selection for specific integers:

- **Ring Stability:** 5- and 6-membered carbon rings minimize angle and torsional strain (e.g., ribose, benzene). 4-membered rings (high angle strain) and 7-membered rings (entropic cost) are rare in metabolic pathways.
- **Fullerenes:** C_{60} is the smallest fullerene satisfying the Isolated Pentagon Rule (IPR). C_{58} and C_{62} require adjacent pentagons, resulting in high reactivity and instability.

14.3 Tri-Level Model Justification

The proposed model mirrors this triadic necessity:

Component	Function	Failure Mode (if removed)
$\|A\|$ (Substrate)	Bound	Unbounded bloat (Unphysical)
$\|B(A)\|$ (Potential)	Gradient	Stasis (No drive)
$C(A)$ (Cost)	Selection	Triviality (Shortest path only)

- **Two Dimensions (B, C):** Solvable optimization. The agent follows the gradient with minimal cost.
- **Three Dimensions (A, B, C):** Navigation. The agent must balance expansion against existence constraints within a cost manifold.

14.4 Structural Postulate

> Intelligence requires three irreducible, competing objectives. Fewer dimensions collapse to solvable optimization or fragile simulation; higher dimensions reduce to effective 3D projections or unstable dynamics.

Relevant Work

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